

1.12 Two Cs

1948

Norbert Wiener, who had been a BA in maths at age 14, published “Cybernetics,” a major influence on later research into AI.

In addition to this, Wiener is noted for such things as his analysis of brain waves, and his exploration of how similar (or dissimilar) the human brain and the computer are—in terms of memory, decision-making, and so on.

For more on cybernetics, which happens to be an important field all its own, visit <http://pespmc1.vub.ac.be/ASC/CYBERNETICS.html>, which has plenty of links.



In addition Think “architecture,” think John von Neumann

Also in 1948, Claude Shannon’s *The Mathematical Theory of Communication* demonstrated to engineers how to code data such that they could check for accuracy after inter-computer transmission. The bytes weren’t there, but the bits were, by this time, and it was now formalised: Shannon identified the bit as the fundamental unit of data. He also (but this happens to be coincidental!) identified it as the basic unit of computation.

This is no particular end-point, but we continue our journey from here into the 50s and on. But most of the foundations had been laid for the computer as we know it: vacuum tubes, then transistors; symbolic logic as bits; computer science; the first few computers.